

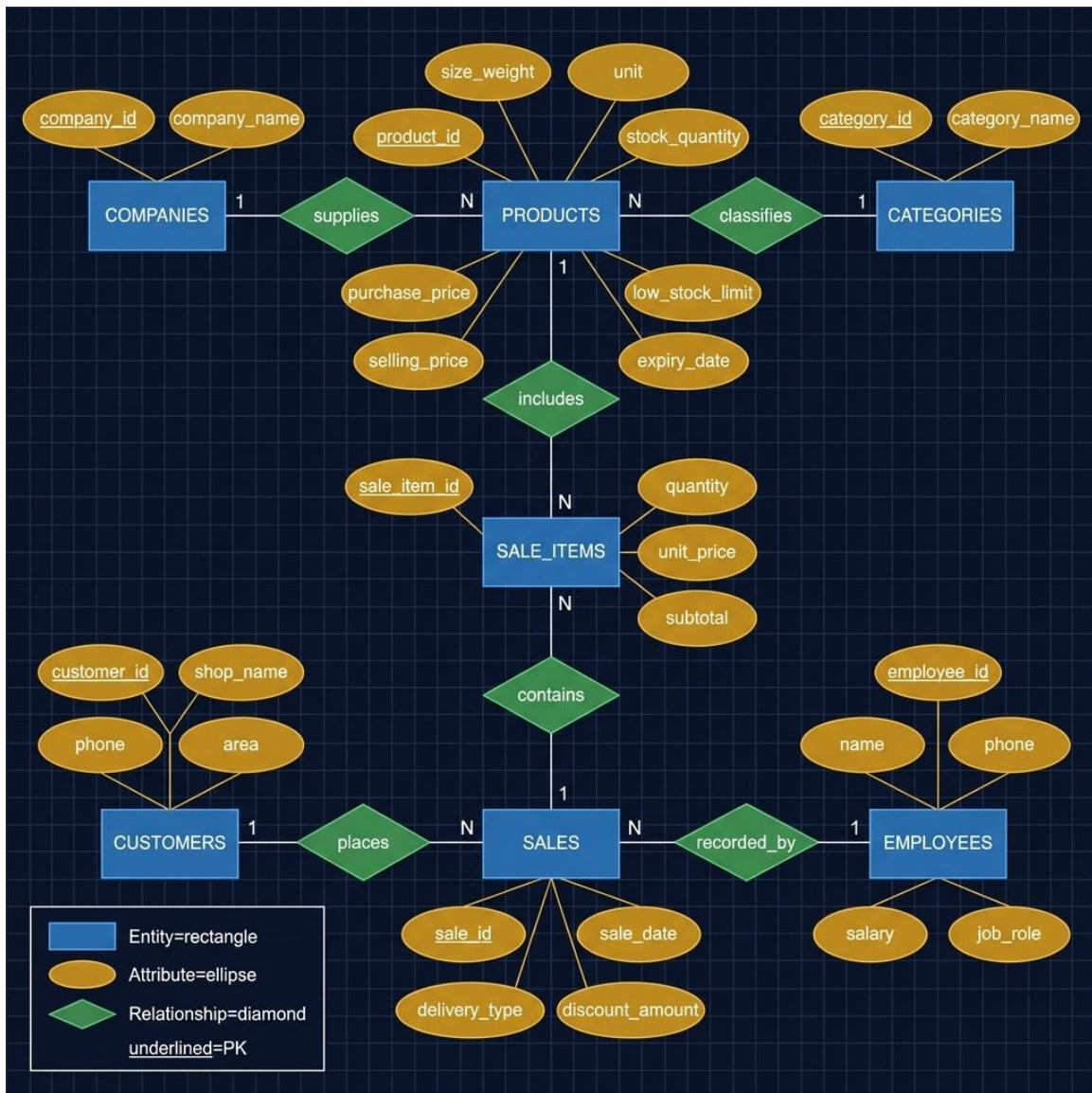
Inventory & Sales Management System

Version Control & Database Lab Submission

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GitHub Repository:	https://github.com/musafkhan100-ops/inventory-sales-management

1. Updated ERD Diagram

Below is our finalized and verified Entity Relationship Diagram layout. It accurately reflects our 3NF database schema, explicitly mapping all primary and foreign key relationships.



2. Normalization Walkthrough (1NF to 3NF)

Original Schema (from Milestone 1)

Entity	Attributes
COMPANIES	company_id (PK), company_name
CATEGORIES	category_id (PK), category_name
PRODUCTS	product_id (PK), product_name, size_weight, unit, stock_quantity, purchase_price, selling_price, low_stock_limit, expiry_date, company_id (FK), category_id (FK)
CUSTOMERS	customer_id (PK), shop_name, phone, area
EMPLOYEES	employee_id (PK), name, phone, salary, job_role
SALES	sale_id (PK), sale_date, delivery_type, discount_amount, total_amount, customer_id (FK), employee_id (FK)
SALE_ITEMS	sale_item_id (PK), quantity, unit_price, subtotal, sale_id (FK), product_id (FK)

First Normal Form (1NF) Analysis

Rule: All columns must contain atomic values, each row must be unique (via PK), and no repeating groups.

- COMPANIES, CATEGORIES, CUSTOMERS, EMPLOYEES: All contain single atomic attributes. Unique PKs. No repeating groups. (✓ Already in 1NF)
- PRODUCTS: All attributes store single atomic values. Product items are not repeated — line items are stored separately in SALE_ITEMS. (✓ Already in 1NF)
- SALES & SALE_ITEMS: All attributes are atomic. Products sold are stored in SALE_ITEMS (no repeating groups here). (✓ Already in 1NF)

Conclusion: All tables satisfy 1NF. The separation of line items into SALE_ITEMS was a key design decision that prevents repeating groups.

Second Normal Form (2NF) Analysis

Rule: Must be in 1NF + no partial dependencies (non-key attributes must depend on the entire primary key).

Key Observation: All seven tables use single-column primary keys (auto-increment integers). Partial dependencies can only occur with composite keys. Therefore, 2NF is automatically satisfied.

Conclusion: All tables satisfy 2NF due to single-column surrogate primary keys. No composite PKs are used, making partial dependencies impossible.

Third Normal Form (3NF) Analysis

Rule: Must be in 2NF + no transitive dependencies (non-key attributes must not depend on other non-key attributes).

- COMPANIES, CATEGORIES, CUSTOMERS, EMPLOYEES, PRODUCTS: Only independent attributes depending directly on the PK. Transitive dependency is impossible or non-existent. (✓ Already in 3NF)

- SALES: ⚠ VIOLATION: total_amount is a derived column. It can be computed as $\text{SUM}(\text{sale_items.quantity} * \text{sale_items.unit_price}) - \text{discount_amount}$. It transitively depends on SALE_ITEMS data.

- SALE_ITEMS: ⚠ VIOLATION: subtotal is a derived column. It equals $\text{quantity} * \text{unit_price}$. It depends on two other non-key attributes, not directly on sale_item_id.

Changes Made for 3NF:

1. SALES: Removed total_amount — it will be computed via SQL query when needed.
2. SALE_ITEMS: Removed subtotal — it will be computed as $\text{quantity} * \text{unit_price}$.

3. Dataflow Description

1. Where Data Enters the System

Data enters our system through two primary channels:

- **Administrative/Managerial Input:** Store managers or admins manually input master data into the system. This includes registering new supplier COMPANIES, defining product CATEGORIES, adding new PRODUCTS to the inventory, and registering new EMPLOYEES.
- **Transactional Input:** Data enters continuously during daily operations when CUSTOMERS place orders at the shop. The EMPLOYEES (cashiers/sales reps) process these transactions at the point of sale.

2. How Data Moves Through the Database

The movement of data relies heavily on the relationships defined in our normalized schema:

- **Master Data Dependency:** Before any sale can occur, the PRODUCTS table must be populated. The PRODUCTS table itself depends on the COMPANIES and CATEGORIES tables being populated first.
- **Transaction Flow:** When a customer makes a purchase, their information is first verified or added to the CUSTOMERS table. A new record is then generated in the SALES table, capturing the sale_date, delivery_type, and any discount_amount. This SALES record links directly to the customer_id and the employee_id.
- **Line Item Resolution:** For every individual product in that sale, a record is created in the SALE_ITEMS table. This table acts as the critical junction, linking the sale_id to the specific product_id, and recording the quantity and historical unit_price.

3. What Comes Out of the System

The structured data flowing into the database is extracted and transformed into actionable insights and reports:

- **Inventory Alerts & Reports:** By querying the PRODUCTS table where stock_quantity <= low_stock_limit, the system outputs automated low-stock alerts.
- **Financial & Sales Reports:** Querying SALES and SALE_ITEMS generates daily/monthly revenue reports, profit margin analysis, and identifies top-selling product categories.
- **Performance & CRM Output:** Data is output as employee performance reports and customer purchase history reports for targeted promotions.